Setup Guide: Julia, JuMP and Gurobi (macOS & Windows)

Follow these steps to install **Julia**, the **JuMP** modeling package and the **Gurobi** solver, and to run Julia in Jupyter notebooks. Allow about 30 minutes.

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1. Install Julia

macOS

1. Open **Terminal** and run:
   ```
   curl -fsSL https://install.julialang.org | sh
   ```
2. If it asks how to install, press **Enter** to accept the default.
3. Close and reopen Terminal, then check that Julia is installed:
   ```
   julia --version
   ```

Windows

1. Open **PowerShell** and run:
   ```
   winget install --name Julia --id 9NJNWW8PVKMN -e -s msstore
   ```
2. Close and reopen PowerShell, then check that Julia is installed:
   ```
   julia --version
   ```

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2. Install the Julia packages

1. Type `julia` in Terminal or PowerShell to start Julia.
2. Press `]` to enter package mode. The prompt changes to `pkg>`.
3. Run:
   ```
   add JuMP Gurobi HiGHS CSV DataFrames Plots IJulia
   ```
4. Wait for it to finish. The first time can take 5–15 minutes.
5. Press **Backspace** to leave package mode.

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3. Get a Gurobi academic license

1. Create a free account at gurobi.com with your **NYU email**, and choose **Academic** as the account type.
2. Download and install the **Gurobi Optimizer** from the Gurobi download page.
3. Log in to the Gurobi User Portal and go to **Licenses → Request**. Choose **Named-User Academic** and click **Generate Now**.
4. Copy the command the portal shows. It looks like this:
   ```
   grbgetkey xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
   ```
5. Connect to **NYU Wi-Fi** or the **NYU VPN**. Gurobi checks this when you activate the license.

6. Open a new Terminal (macOS) or PowerShell (Windows), paste the command and press **Enter**.
7. When it asks where to save the license, press **Enter** to keep the default (your home folder).

You do not need to set `GUROBI_HOME` or change your PATH. The license lasts one year; to renew it, repeat steps 3–7.

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4. Test your setup

1. Start Julia and run:

```
using JuMP, Gurobi
model = Model(Gurobi.Optimizer)
@variable(model, x >= 0)
@variable(model, y >= 0)
@constraint(model, x + y <= 10)
@objective(model, Max, 5x + 3y)
optimize!(model)
println(termination_status(model))
println(objective_value(model))
```

2. You should see a line saying **Academic license** with its expiry date, and these two lines at the end:

```
OPTIMAL
50.0
```

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5. Use Julia in Jupyter

1. Start Jupyter the way you normally do, for example:

```
jupyter notebook
```

2. Choose the **Julia** kernel when you create or open a notebook.

If you don't have Jupyter installed, start Julia and run the lines below. The first time, this installs Jupyter and then opens it in your browser.

```
using IJulia
notebook()
```

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6. Troubleshooting

- **"No Gurobi license found":** check that `gurobi.lic` is in your home folder.
- **grbgetkey is not recognized:** close and reopen Terminal or PowerShell after installing Gurobi.
- **Error about your academic domain or network:** connect to NYU Wi-Fi or the NYU VPN, generate a new license in the portal, and try again.
- **The Julia kernel is missing in Jupyter:** in Julia, run `using Pkg; Pkg.build("IJulia")`, then restart Jupyter.
- **"Package Gurobi not found":** repeat step 2.

You are now ready to solve optimization problems in Julia using Gurobi!